

The empirical recurrence for OEIS A233017: recovering a conjecture that is not written down

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Abstract

OEIS A233017 records “Empirical recurrence of order 33”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 71$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 33, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 33 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233017 is “Number of $n \times 7$ 0..3 arrays with no element $x(i,j)$ adjacent to value $3-x(i,j)$ horizontally or vertically, top left element zero, and 1 appearing before 2 in row major order.”. It has offset 1 and begins

365, 176474, 85827722, 41775098873, 20336458202315, 9900284810558324, 4819736584585663322, 2346387082638

The entry states:

Empirical recurrence of order 33 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:47:02, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 71$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 142$ exact terms of the model returns order 33 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{33} c_i a(n-i),$$

c_1	820	c_{10}	−63890727017432669	c_{19}	−5200219344858789634094007	c_{28}	−8
c_2	−205689	c_{11}	4141584863538946248	c_{20}	79782056692975632581941806	c_{29}	77
c_3	23977335	c_{12}	−41819886892303325457	c_{21}	−145976694071885459756226729	c_{30}	−20
c_4	−1448890080	c_{13}	−30483804619910407683	c_{22}	−381423099164490179830401489	c_{31}	22
c_5	40681354716	c_{14}	3863177257558021068078	c_{23}	1687752294947845021038392100	c_{32}	6
c_6	113470351405	c_{15}	−27574675558592643871665	c_{24}	−947863788801858331624770840	c_{33}	—
c_7	−45083135350759	c_{16}	−5974253069771812381395	c_{25}	−4416578720165084280803867511		
c_8	1384281526165416	c_{17}	994204560645628682891988	c_{26}	7519754083669273956859949007		
c_9	−16268081626571563	c_{18}	−4133294712546191129276631	c_{27}	−393515001351016977961508328		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 33, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $33 + 4$ terms removed returns order 33 as well, so $\deg q_{\min} = 33 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $33 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 33 that this sequence satisfies — and that family turns out to have exactly one member.

References

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